

Week 6 Lab Task

Network Sniffing & WiFi Hacking

PART 1 — Network Sniffing (Diagnose Connectivity Issues)

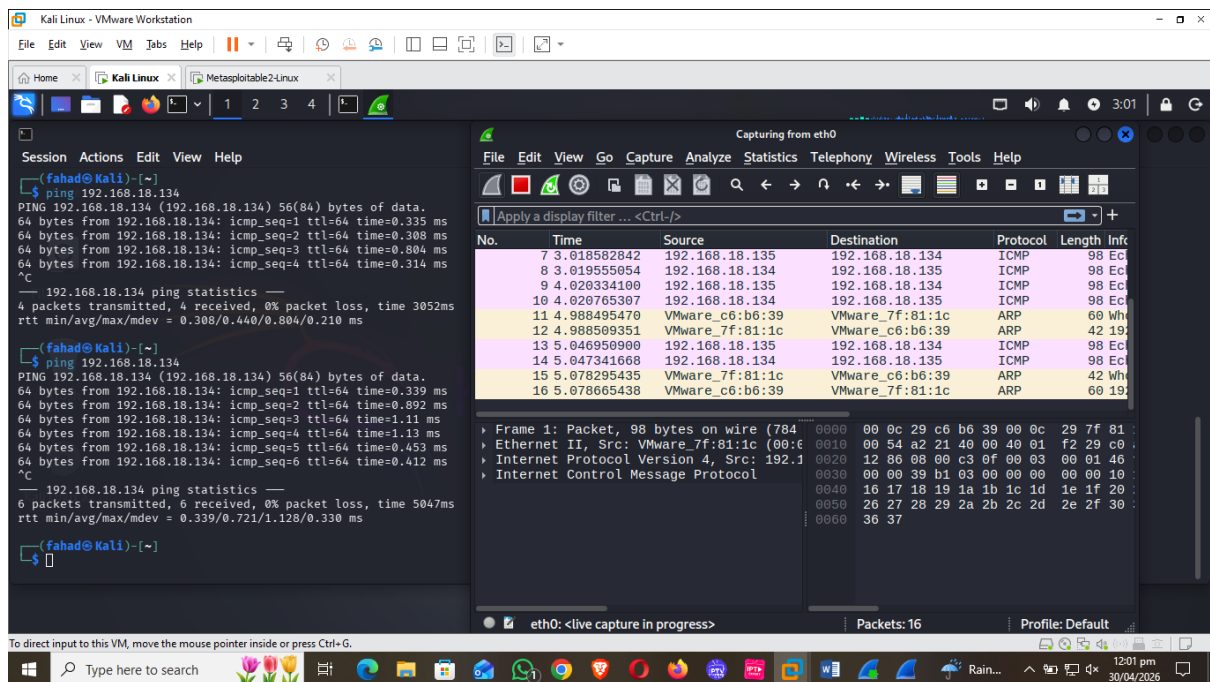
Step 1 — Configure the Lab Network

In VirtualBox / VMware

Make sure:

- Kali Linux = Host-Only Adapter
- Metasploitable 2 = Same Host-Only Adapter

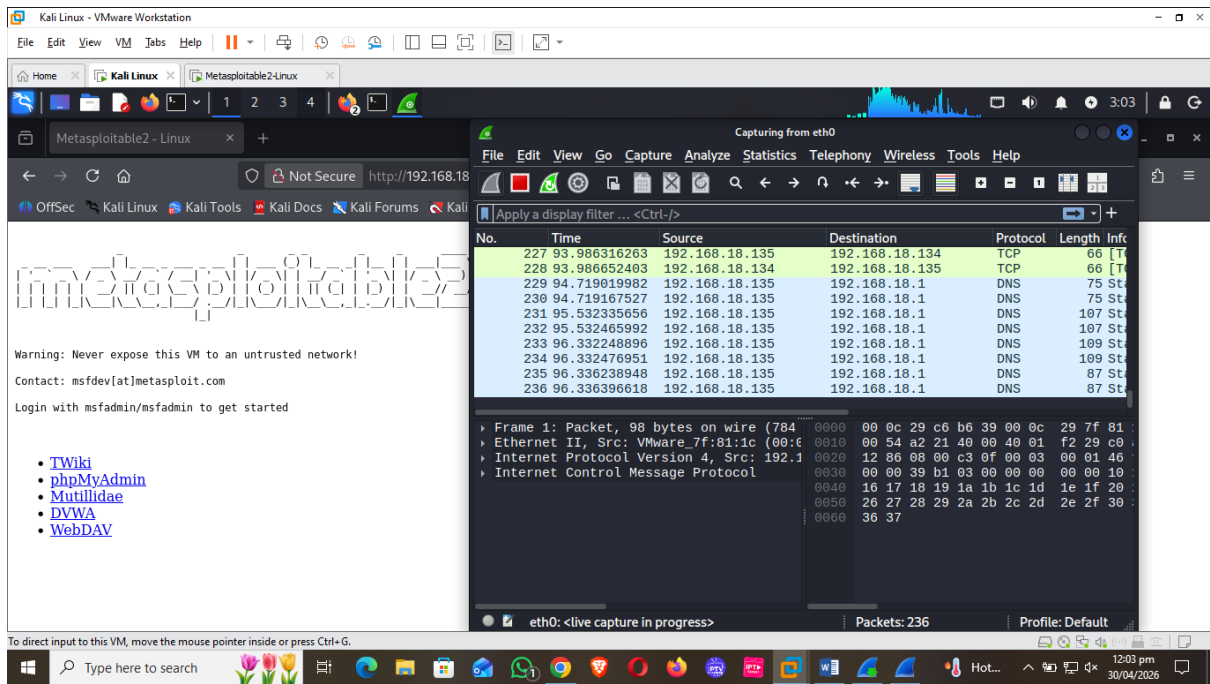
Both VMs must be on same subnet.

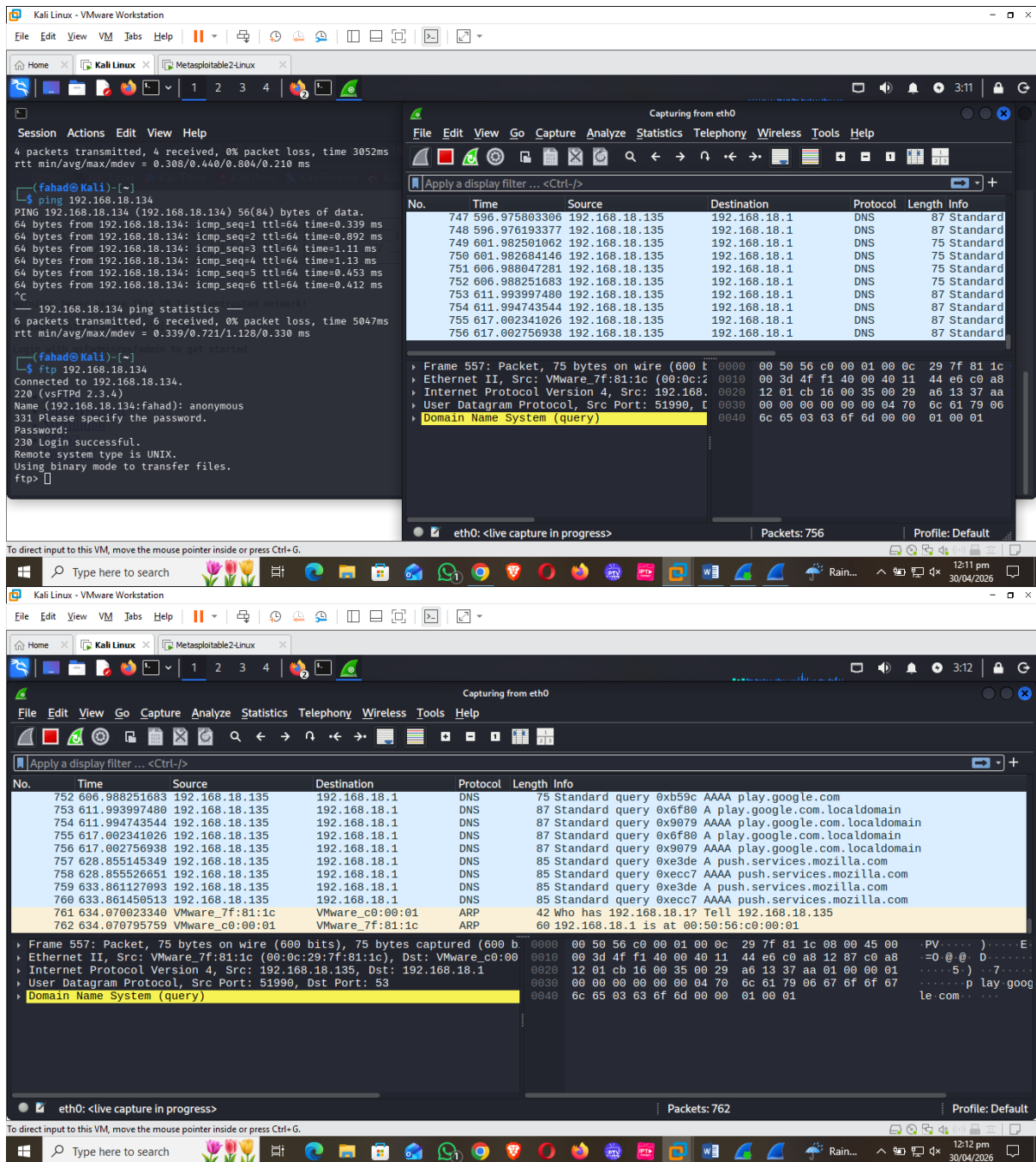


Step 2 — Capture Traffic Using Wireshark

Open Wireshark

Applications → Sniffing & Spoofing → Wireshark

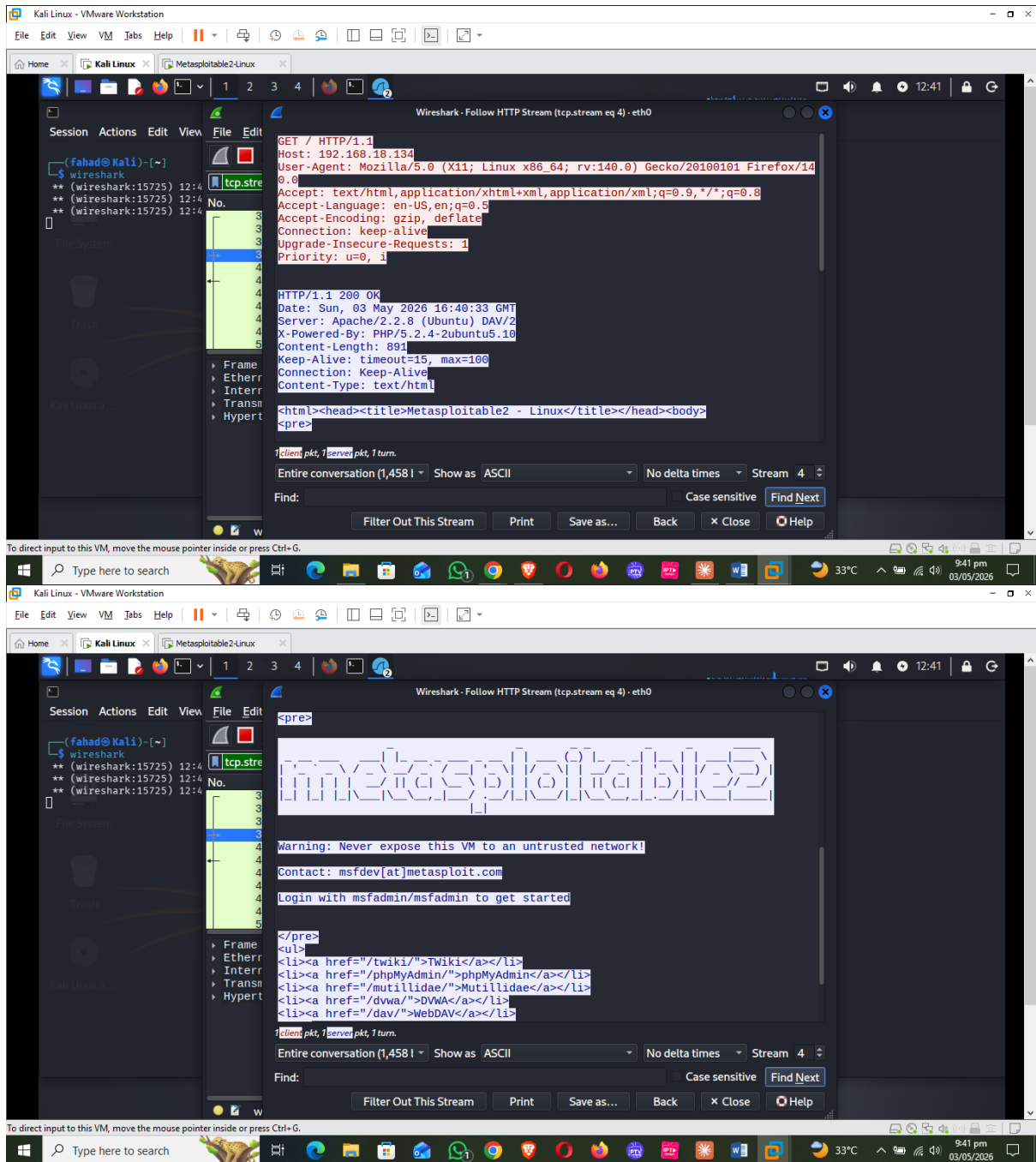




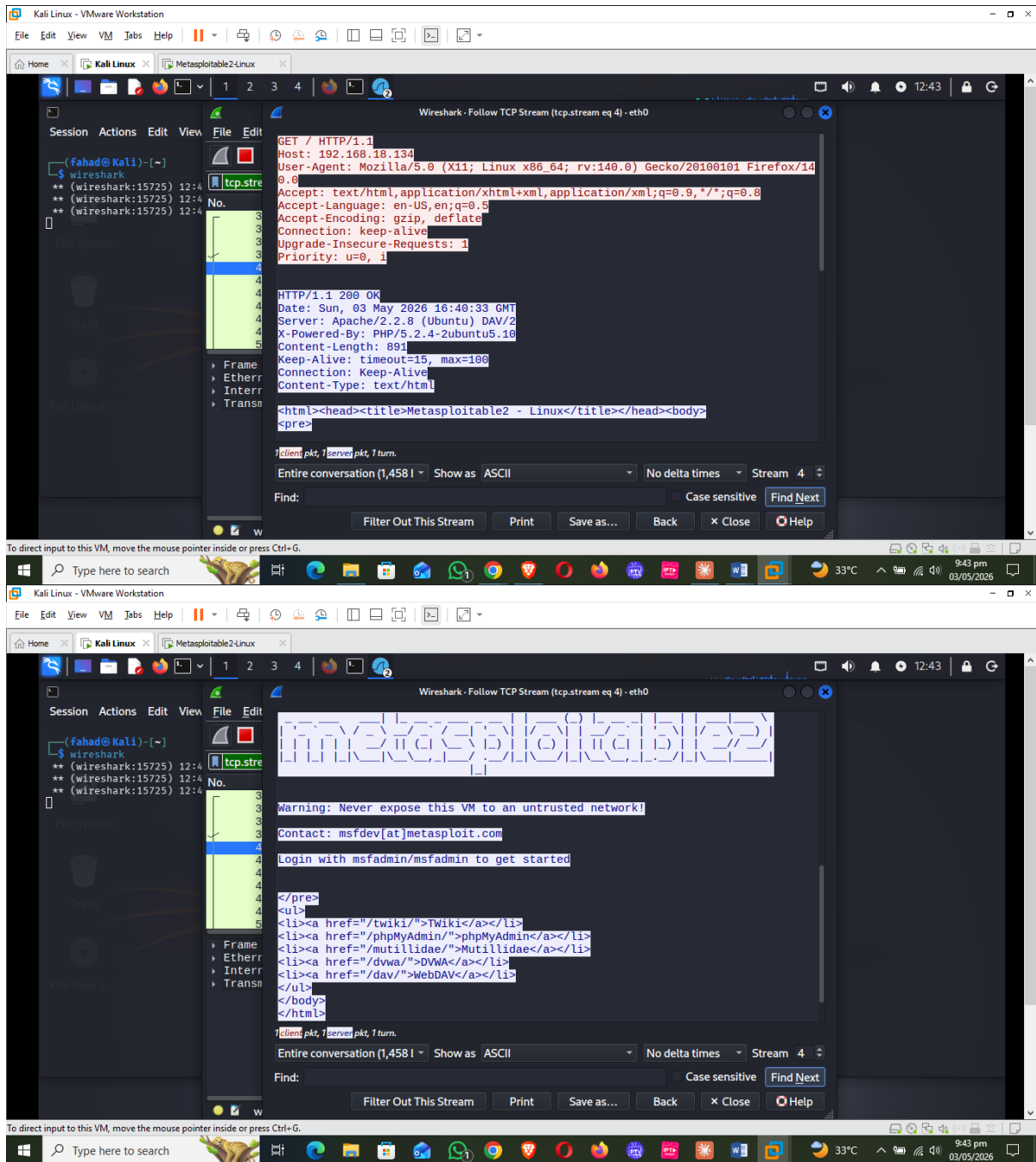
Step 3 — Analyze Traffic in Wireshark

Useful Filters

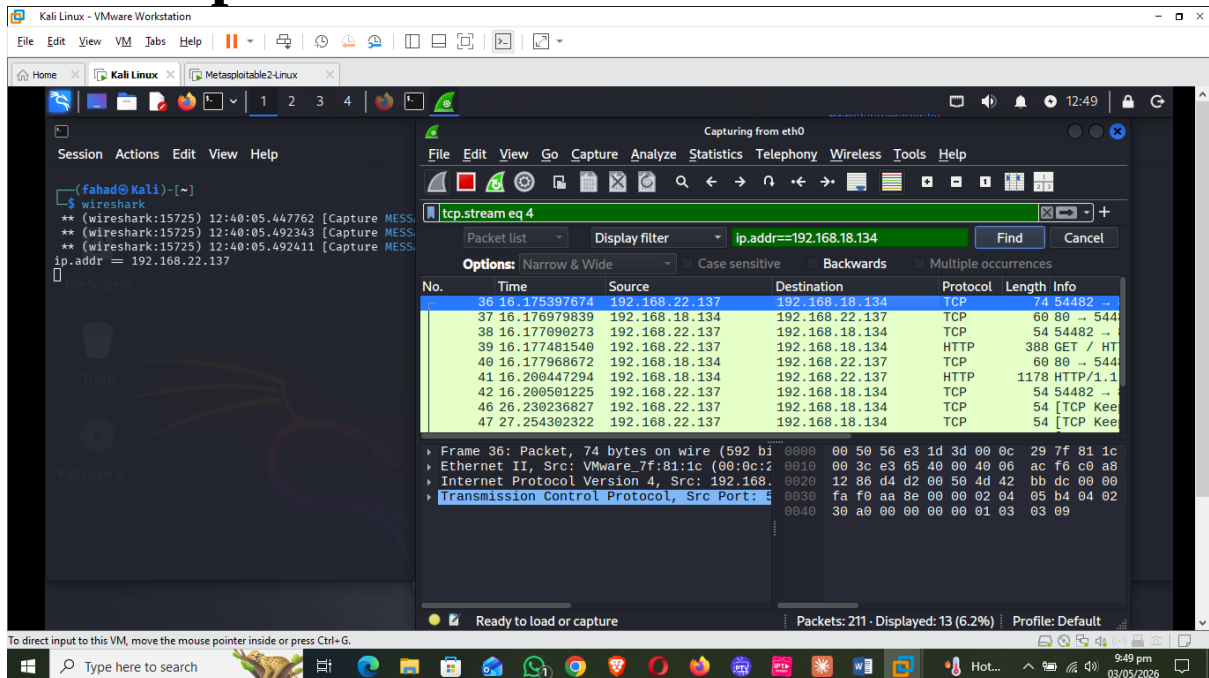
HTTP Traffic:



TCP Packets:

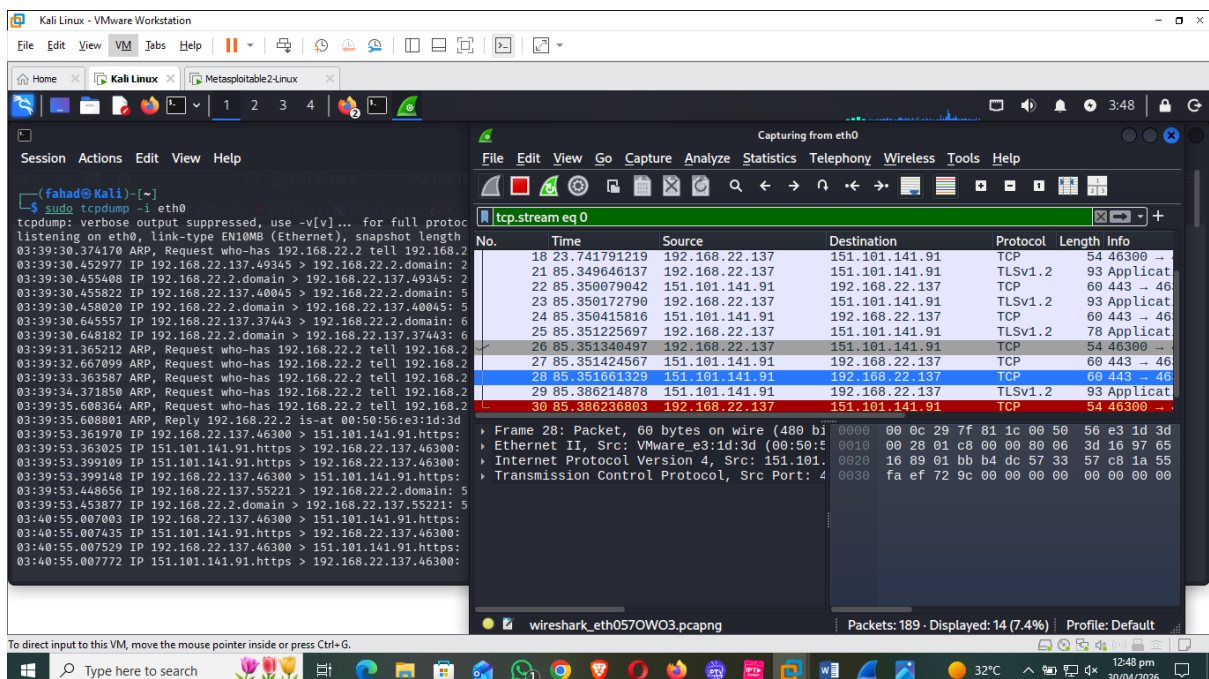


Filter Specific IP:



Step 4 — Capture Traffic with tcpdump

Capture Everything



Save Capture File:

The screenshot shows a Kali Linux virtual machine running on VMware Workstation. The terminal window displays the output of a tcpdump command, showing a list of ARP requests and replies between 192.168.22.2 and 192.168.22.1. The Wireshark window is open, capturing traffic on the eth0 interface. The packet list shows a series of ARP requests and replies. The packet details pane shows the structure of an ARP request, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol.

Session Actions Edit View Help

```
03:46:24.363098 ARP, Request who-has 192.168.22.2 tell 192.168.22.1, length 46
03:46:25.390289 ARP, Request who-has 192.168.22.2 tell 192.168.22.1, length 46
03:46:27.769425 ARP, Request who-has 192.168.22.2 tell 192.168.22.137, length 28
03:46:27.771103 ARP, Reply 192.168.22.2 is-at 00:50:56:e3:1d:3d (oui Unknown), length 46
03:46:29.652928 ARP, Request who-has 192.168.22.2 tell 192.168.22.1, length 46
03:46:30.376810 ARP, Request who-has 192.168.22.2 tell 192.168.22.1, length 46
03:46:31.377348 ARP, Request who-has 192.168.22.2 tell 192.168.22.1, length 46
03:46:32.671008 ARP, Request who-has 192.168.22.2 tell 192.168.22.1, length 46
03:46:33.363459 ARP, Request who-has 192.168.22.2 tell 192.168.22.1, length 46
03:46:34.367630 ARP, Request who-has 192.168.22.2 tell 192.168.22.1, length 46
03:48:11.864769 ARP, Request who-has 192.168.22.2 tell 192.168.22.1, length 46
03:48:13.089666 ARP, Request who-has 192.168.22.2 tell 192.168.22.1, length 46
03:48:13.063267 ARP, Request who-has 192.168.22.2 tell 192.168.22.1, length 46
03:48:14.874893 ARP, Request who-has 192.168.22.2 tell 192.168.22.1, length 46
03:48:20.623512 ARP, Request who-has 192.168.22.2 tell 192.168.22.1, length 46
03:48:21.364548 ARP, Request who-has 192.168.22.2 tell 192.168.22.1, length 46
03:48:22.370420 ARP, Request who-has 192.168.22.2 tell 192.168.22.1, length 46
^C
179 packets captured
180 packets received by filter
0 packets dropped by kernel

(fahad@Kali)-[~]
$ sudo tcpdump -i eth0 -w capture.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

Capturing from eth0

No.	Time	Source	Destination	Protocol
21	85.349646137	192.168.22.137	151.101.141.91	T
22	85.350079042	151.101.141.91	192.168.22.137	T
23	85.350172790	192.168.22.137	151.101.141.91	T
24	85.350415816	151.101.141.91	192.168.22.137	T
25	85.351225697	192.168.22.137	151.101.141.91	T
26	85.351340497	192.168.22.137	151.101.141.91	T
27	85.351424567	151.101.141.91	192.168.22.137	T
28	85.351661329	151.101.141.91	192.168.22.137	T
29	85.386214878	151.101.141.91	192.168.22.137	T
30	85.386236803	192.168.22.137	151.101.141.91	T

Frame 28: Packet, 60 bytes on wire (480 bits) captured (0x00000000) on interface eth0
Ethernet II, Src: VMware_e3:1d:3d (08:00:50:56:e3:1d), Dst: 151.101.141.91 (08:00:50:56:e3:1d)
Internet Protocol Version 4, Src: 151.101.141.91, Dst: 192.168.22.137
Transmission Control Protocol, Src Port: 443, Dst Port: 80

wireshark_eth057OWO3.pcapng

Packets: 206 · Displayed: 14

Filter Specific Host:

The screenshot shows a Kali Linux virtual machine running on VMware Workstation. The terminal window displays the output of a tcpdump command, showing a list of IP addresses and their corresponding domain names. The Wireshark window is open, capturing traffic on the eth0 interface. The packet list shows a series of IP addresses and their corresponding domain names. The packet details pane shows the structure of an IP packet, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol.

Session Actions Edit View Help

```
0 packets dropped by kernel

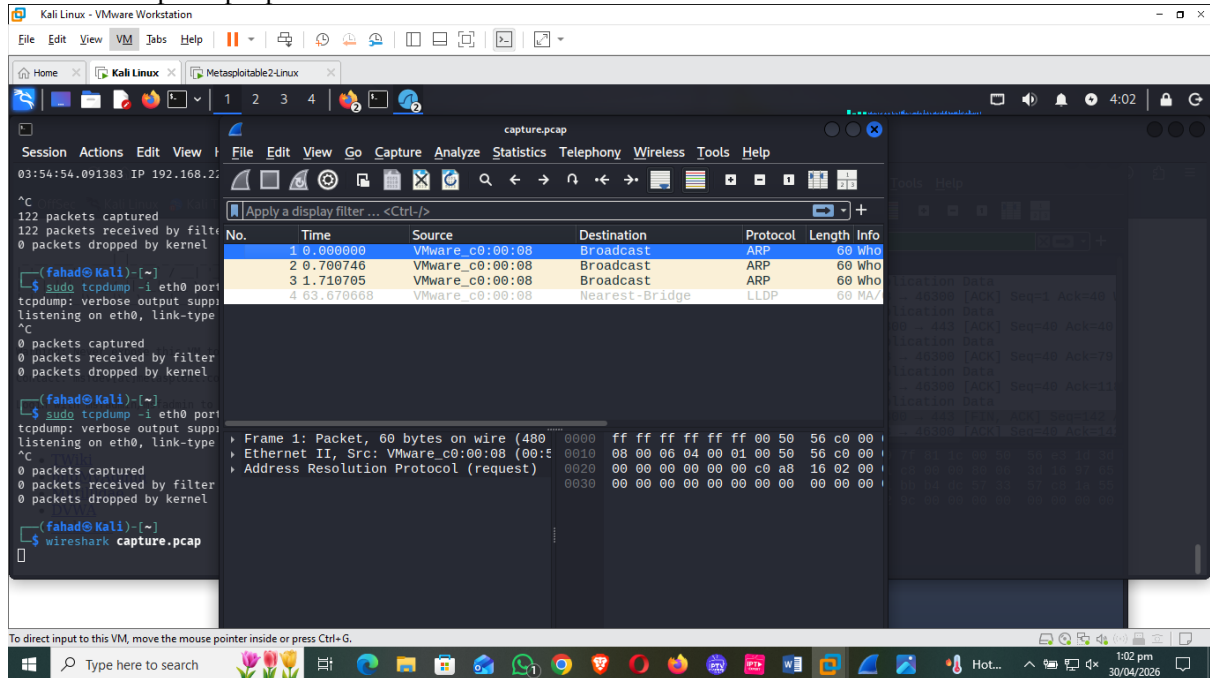
(fahad@Kali)-[~]
$ sudo tcpdump -i eth0 host 192.168.22.137
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
03:54:25.506968 IP 192.168.22.137.60139 > 192.168.22.2.domain: 24144+ A? safebrowsing.googleapis.com. (45)
03:54:25.507247 IP 192.168.22.137.60139 > 192.168.22.2.domain: 56747+ AAAA? safebrowsing.googleapis.com. (45)
03:54:25.509775 IP 192.168.22.2.domain > 192.168.22.137.60139: 24144 1/0/0 A 142.250.200.170 (61)
03:54:25.512504 IP 192.168.22.2.domain > 192.168.22.137.60139: 56747 1/0/0 AAAA 2400:1450:4019:80f::200a (73)
03:54:25.514633 IP 192.168.22.137.34738 > pnfjra-aq-in-f10.1e100.net.https: Flags [S], seq 1277165463, win 64240, options [mss 1460,sackOK,TS val 834466570 ecr 0,nop,wscale 9], length 0
03:54:25.551667 IP pnfjra-aq-in-f10.1e100.net.https > 192.168.22.137.34738: Flags [S.], seq 386352984, ack 1277165464, win 64240, options [mss 1460], length 0
03:54:25.551739 IP 192.168.22.137.34738 > pnfjra-aq-in-f10.1e100.net.https: Flags [.], ack 1, win 64240, length 0
03:54:25.553265 IP 192.168.22.137.34738 > pnfjra-aq-in-f10.1e100.net.https: Flags [P.], seq 1:1913, ack 1, win 64240, length 1912
03:54:25.553585 IP pnfjra-aq-in-f10.1e100.net.https > 192.168.22.137.34738: Flags [.], ack 1461, win 64240, length 0
03:54:25.553585 IP pnfjra-aq-in-f10.1e100.net.https > 192.168.22.137.34738: Flags [.], ack 1913, win 64240, length 0
03:54:25.568960 IP 192.168.22.137.52523 > 192.168.22.2.domain: 43716+ PTR? 2.22.168.192.in-addr.arpa. (43)
03:54:25.571074 IP 192.168.22.2.domain > 192.168.22.137.52523: 43716 1/0/0 PTR 192.168.22.2. (60)
03:54:25.571389 IP 192.168.22.137.48178 > 192.168.22.2.domain: 30186+ PTR? 137.22.168.192.in-addr.arpa. (45)
03:54:25.573164 IP 192.168.22.2.domain > 192.168.22.137.48178: 30186 1/0/0 PTR 192.168.22.137. (73)
03:54:25.573568 IP 192.168.22.137.37449 > 192.168.22.2.domain: 2647+ PTR? 170.200.250.142.in-addr.arpa. (46)
03:54:25.577669 IP 192.168.22.2.domain > 192.168.22.137.37449: 2647 1/0/0 PTR pnfjra-aq-in-f10.1e100.net. (86)
03:54:25.743272 IP pnfjra-aq-in-f10.1e100.net.https > 192.168.22.137.34738: Flags [P.], seq 1:5598, ack 1913, win 64240, length 5597
03:54:25.743335 IP 192.168.22.137.34738 > pnfjra-aq-in-f10.1e100.net.https: Flags [.], ack 5598, win 65535, length 0
03:54:25.754160 IP 192.168.22.137.34738 > pnfjra-aq-in-f10.1e100.net.https: Flags [P.], seq 1913:1993, ack 5598, win 65535, length 80
03:54:25.754644 IP pnfjra-aq-in-f10.1e100.net.https > 192.168.22.137.34738: Flags [.], ack 1993, win 64240, length 0
03:54:25.755387 IP 192.168.22.137.34738 > pnfjra-aq-in-f10.1e100.net.https: Flags [P.], seq 1993:2085, ack 5598, win 65535, length 92
```

CPU usage: 2.1%

wireshark_eth057OWO3.pcapng

Packets: 387 · Displayed: 14 (3.6%)

wireshark capture.pcap:



Protocols Observed

- ARP
- ICMP
- TCP
- HTTP
- FTP
- DNS

Sensitive Data Exposure

Observed plaintext:

- FTP credentials
- HTTP requests
- Telnet session data

Risk:

Attackers can steal credentials using sniffing.

Root Cause of Connectivity Problem

Possible findings:

- TCP retransmissions causing delays
- DNS misconfiguration
- Packet loss

- Excessive broadcast traffic
- Unencrypted protocols

Recommended Solutions

Security Improvements

- Replace HTTP with HTTPS
- Replace Telnet with SSH
- Use secure FTP alternatives (SFTP)